

# Myanmar Earthquake & Dam Risk Report

## 1. Executive Summary

This report covers 9,403 earthquake events in Myanmar from 1970-07-29 to 2026-06-30. The largest event was M7.7 on 1988-11-06. The Gutenberg-Richter b-value is 1.049 with magnitude of completeness  $M_c=4.7$ .

**Total Events:** 9,403

**Date Range:** 1970-07-29 to 2026-06-30

**Max Magnitude:** M7.7

**Mean Depth:** 24.8 km

**b-value:** 1.049

**a-value:** 7.715

**$M_c$  (completeness):** 4.7

## 2. Earthquake Map

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## 3. Seismic Clustering Analysis

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## 4. Dam Seismic Risk Assessment

Risk assessment for 254 dams using PGA estimation from the M7.7 mainshock, fault proximity, and structural exposure scoring.

**Critical Risk:** 25 dams

**High Risk:** 148 dams

**Moderate Risk:** 67 dams

**Low Risk:** 14 dams

### Top 30 Highest Risk Dams:

| Name                   | River          | PGA(g)  | Dist    | Fault    | Risk     | Grade |  |  |  |
|------------------------|----------------|---------|---------|----------|----------|-------|--|--|--|
| Dam near 22.09N 95.86E |                | 0.4019  | 5.7 km  | 8.3      | Critical |       |  |  |  |
| Dam near 21.77N 95.89E |                | 0.4626  | 2.7 km  | 8.3      | Critical |       |  |  |  |
| Natkar                 | 0.3593         | 7.9 km  | 8.2     | Critical |          |       |  |  |  |
| Dam near 21.47N 95.87E |                | 0.3944  | 6.0 km  | 8.2      | Critical |       |  |  |  |
| Ta Nai Hka             | Ta Nai Hka     | 0.4441  | 3.9 km  | 8.2      | Critical |       |  |  |  |
| Dam near 20.72N 95.90E |                | 0.4000  | 5.6 km  | 8.2      | Critical |       |  |  |  |
| Dam near 21.74N 95.85E |                | 0.3656  | 7.4 km  | 8.2      | Critical |       |  |  |  |
| Taungnyo               | Taungnyo       | 0.5047  | 1.2 km  | 8.2      | Critical |       |  |  |  |
| Dam near 20.60N 95.90E |                | 0.3681  | 7.4 km  | 8.2      | Critical |       |  |  |  |
| Dam near 20.47N 96.08E |                | 0.3252  | 9.9 km  | 8.2      | Critical |       |  |  |  |
| Dam near 20.85N 96.03E |                | 0.3478  | 8.3 km  | 8.2      | Critical |       |  |  |  |
| Meikhtila              | 0.3525         | 8.2 km  | 8.2     | Critical |          |       |  |  |  |
| Dam near 17.55N 96.37E |                | 0.3298  | 9.6 km  | 8.1      | Critical |       |  |  |  |
| Dam near 21.05N 95.83E |                | 0.3105  | 11.0 km | 8.0      | Critical |       |  |  |  |
| Dam near 21.04N 95.83E |                | 0.3122  | 10.9 km | 8.0      | Critical |       |  |  |  |
| Thaphan Chaung         | Thaphan Chaung | 0.3041  | 11.5 km | 7.8      | Critical |       |  |  |  |
| Dam near 21.34N 95.81E |                | 0.2956  | 12.2 km | 7.7      | Critical |       |  |  |  |
| Kun Chaung             | Kun Chaung     | 0.5304  | 0.9 km  | 7.6      | Critical |       |  |  |  |
| Phyu Chaung            | Pyu Chaung     | 0.5211  | 1.3 km  | 7.5      | Critical |       |  |  |  |
| Kabaung                | Kabaung Chaung | 0.3302  | 9.2 km  | 7.5      | Critical |       |  |  |  |
| Dam near 21.37N 95.80E |                | 0.2830  | 13.4 km | 7.5      | Critical |       |  |  |  |
| Yenwe                  | Ye Nwe Chaung  | 0.4562  | 3.9 km  | 7.4      | Critical |       |  |  |  |
| Nyaunggone             | 0.2795         | 13.7 km | 7.4     | Critical |          |       |  |  |  |
| Dam near 20.73N 95.83E |                | 0.2809  | 13.6 km | 7.4      | Critical |       |  |  |  |
| Zeedaw                 | Nyaung Oat     | 0.2633  | 15.3 km | 7.2      | Critical |       |  |  |  |
| Male-Nattaung          | Male-Nattaung  | 0.2327  | 18.6 km | 6.9      | High     |       |  |  |  |
| Kyee Ni                | 0.2362         | 19.4 km | 6.8     | High     |          |       |  |  |  |
| Kintat                 | Mu             | 0.2235  | 21.0 km | 6.7      | High     |       |  |  |  |
| Dam near 20.58N 96.17E |                | 0.2281  | 20.7 km | 6.7      | High     |       |  |  |  |
| Myotha                 | Ku Hta         | 0.2211  | 20.6 km | 6.7      | High     |       |  |  |  |

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## 5. Aftershock Forecast

Modified Omori law parameters fitted to aftershocks of M7.7 mainshock.

**K (productivity):** 4.53

**c (time offset):** 0.100 hours

**p (decay exponent):** 0.932

**Expected M $\geq$ 3 (7d):** 0

**Expected M $\geq$ 3 (30d):** 0

**Expected M $\geq$ 3 (90d):** 0

Note: These are statistical estimates based on the Omori decay law. Actual aftershock rates may vary due to local geological conditions.

## 6. Data Sources

- Earthquake data: mmeq.akze.net API (USGS/ISC sourced)
- Dam data: Open Development Mekong (IFC/WLE), CC BY-SA 4.0
- Fault lines: USGS/Plate boundary data
- PGA estimation: Abrahamson & Silva (2008) NGA-West1 GMPE
- Report generated by mmeq-opendata toolkit